

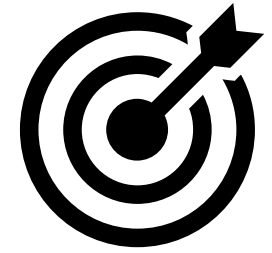


FLIGHT DATA ANALYSIS

**PRESENTED BY: EMILIA PEREZ, FERNANDO
CASILLAS, AHTZIRI AGUIRRE, & ROSSELLY
RINCON**



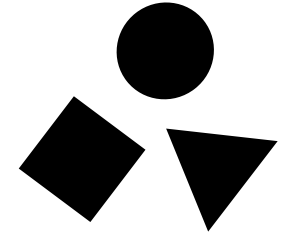
OBJECTIVES



- Upload dataset into ElasticSearch
- Create visualizations using Kibana
- Create machine learning models using the dataset
- Evaluate the performance of the machine learning model

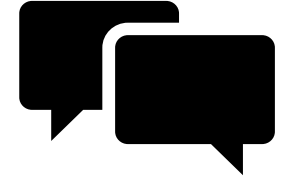


ABSTRACT



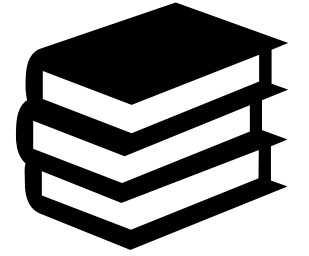
- Used an airline dataset with flight status, routes, passenger demographics, and details.
- Analyzed trends in delays and cancellations by country.
- Identified most frequent travel destinations and the average age of fliers.

INTRODUCTION



- Chosen for its real-world relevance: frequent flight delays and cancellations.
- Recent events (e.g., Spirit Airlines bankruptcy) highlight the need for analysis.
- Goal: examine how often delays or cancellations occur on specific routes and average age of fliers.

RELATED WORK



"Patterns in the Sky: A Flight Delay Analysis"

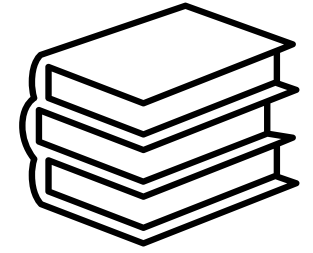
- Performs exploratory data analysis on airline performance data to identify key delay pattern

"Propagation of Delays in the National Airspace System"

- Investigates how flight delays spread across interconnected flights using probabilistic modeling techniques

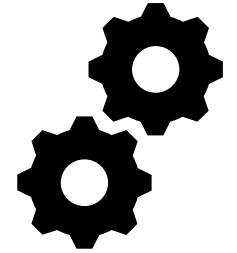
Our work is different due to the use of ElasticSearch machine learning to predict delays.

BACKGROUND/EXISTING WORK



- The Federal Aviation Administration (FAA) predicts flight delays using the Traffic Management Flow System in real time, the Aviation System Performance Metrics, and it monitors the weather 24/7.
- We used Kibana to predict flight delays and cancellations.
- Our data is limited and focuses on flight statuses, flight routes, passenger demographics, and flight details.

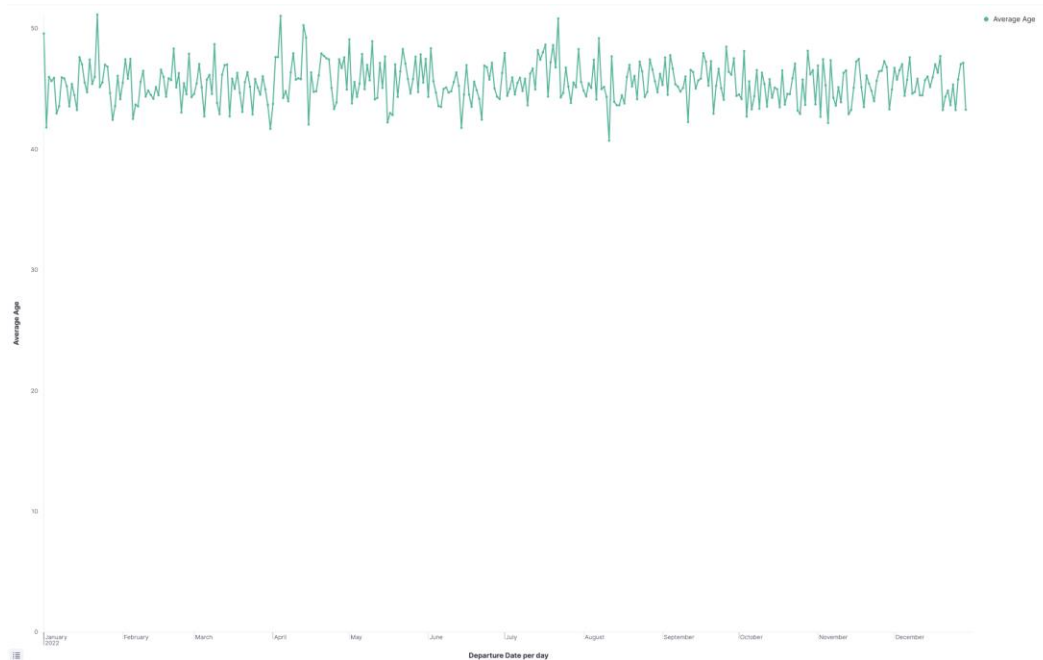
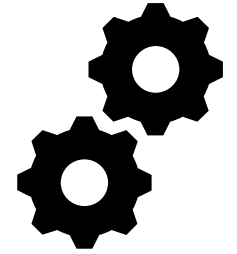
OUR WORK



- Met once a week, discussed designs for the charts and the ideas behind said charts
- Created charts in Kibana to explore data such as average flier age per day
- Created a machine learning model to predict delays in flights

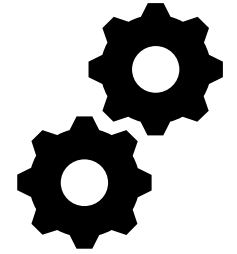


OUR WORK (CONT.)

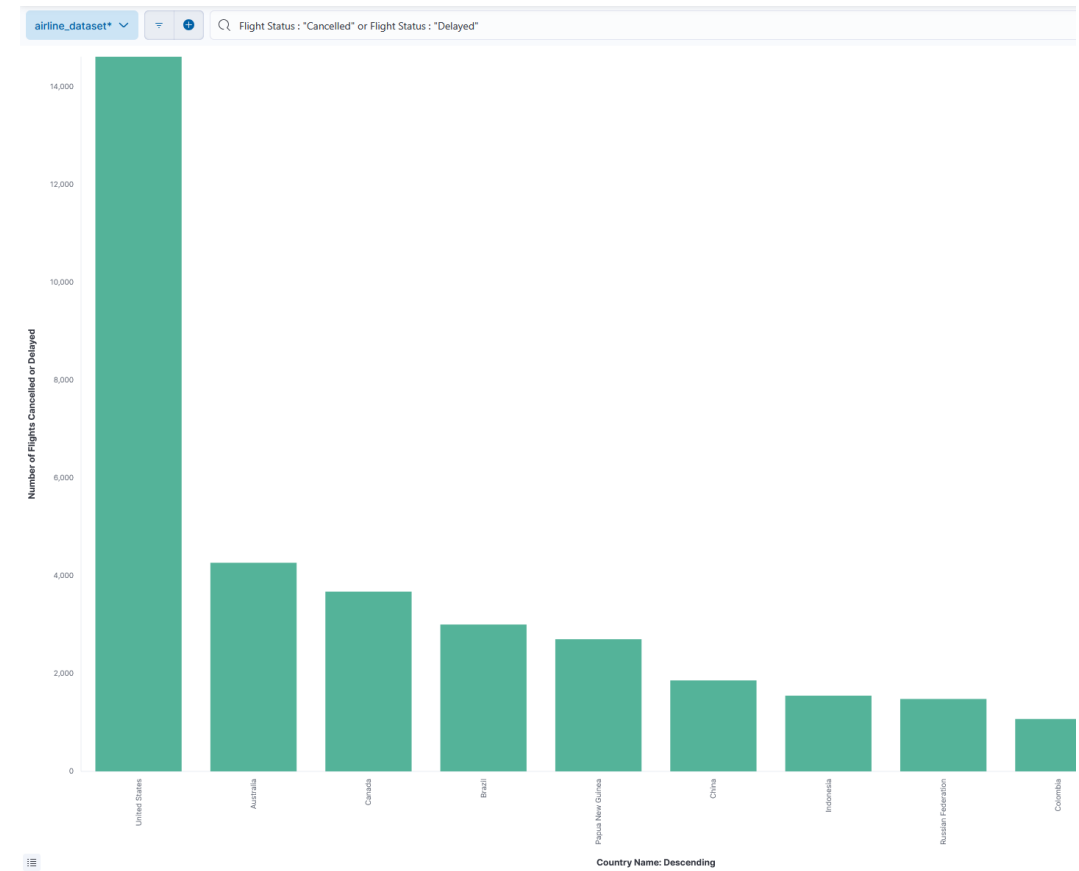


- Multiple charts were created to explore correlations and identify trends in the data.
- This chart shows that the average flier was between 40 and 50 years old for 2022.

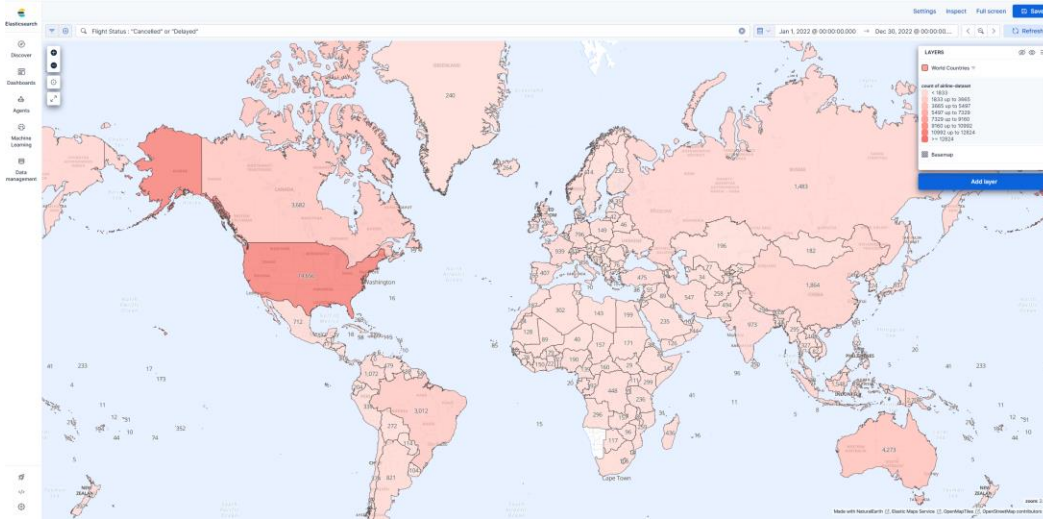
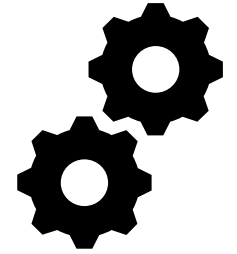
OUR WORK (CONT.)



- The chart focuses on the top ten countries with delays or cancellations
- The United States had significantly more delays or cancellations than any other country



OUR WORK (CONT.)



- Used a map to highlight hotspots of delayed/cancelled flights.
- Helps show which regions are most affected.

OUR WORK (CONT.)



- This classification model predicts flight delays using selected features.
- It achieved 52.4% accuracy with an AUC of 0.53.
- Performance is limited due to weak predictive features.

CONCLUSION

- This project used Elasticsearch and Kibana to analyze airline data and build a delay prediction model
- The classification model achieved 52.4% accuracy, showing the full machine learning process.
- A more comprehensive dataset, including factors like weather or flight operations could improve accuracy.
- The project is unique because it combines data analysis and machine learning within a single Elasticsearch platform.

REFERENCES

Airline Dataset:

<https://www.kaggle.com/datasets/iamsouravbanerjee/airline-dataset>

Github: <https://github.com/StrayBerry512/CIS-3200-Final>

Airline On-Time Statistics and Delay Causes:

https://www.transtats.bts.gov/OT_Delay/OT_DelayCause1.asp

REFERENCES (CONT.)

Ratnayake, Paboda. "Patterns in the Sky: A Flight Delay Analysis." Medium, 18 May 2025, medium.com/@paboda-ratnayake/patterns-in-the-sky-a-flight-delay-analysis-1444df5d82d1

Laskey, Kathryn , et al. "Propagation of Delays in the National Airspace System." arxiv.org/pdf/1206.6859